The POC that is already running

This guide is the first part of the file you send to Ashish and Ananya. It describes the connector stand-in that is already in motaxle-jinchen-connector. They should read it before they change anything. The remaining tasks start on the next part of this file. Those pages are numbered Page 1 again, and their section numbers stay 1 to 13.

ERPNext posts stock and accounts. Jinchen reports what the machine did. The connector is a separate program between them. On this machine the connector does not call Jinchen and it does not call an ERPNext site. Stock is an in-memory ledger. A Jinchen message is a JSON object. The field names in that JSON are the POC contract. They are not a signed Jinchen API.

1. What 'API pack is not signed' means

The official interface note has not arrived and has not been agreed. It is not a missing digital signature on a file. Until that note marks a call in scope, nobody adds a Jinchen URL, a token, or a path copied from another factory. The buttons on the desk are the local stand-in for those messages.

2. Where the code lives

File	What it already does
connector/poc.py	The ledger, the checks, the repeat rule, and heat lookup. This is the stand-in. Do not write a second one.
demo.py	Prints the whole job in the terminal, including the failed log write and the repeat.
tests/test_poc.py	Twelve tests. They lock the figures 380, 384, 96, and 0.
server.py	The local website on 127.0.0.1 port 8765. One desk, shared by every browser tab.
README.md	How to start the desk, the terminal story, and the tests.

Python 3.9 or newer, standard library only. No Flask, no requests, no Frappe. The folder is not inside motaxle-bench/apps. Do not run bench new-app for this connector.

3. How to run what is already there

From the motaxle-jinchen-connector folder:

```
python3 demo.py
python3 -m unittest tests/test_poc.py
python3 server.py
```

The website address is http://127.0.0.1:8765. The port is required. 127.0.0.1 alone is not this desk, and http://motaxle.localhost:8000 is the ERPNext desk, which is not this POC. Ctrl+C stops the server. It does not delete the folder or the site. Reset job on the page puts the ledger back to 500 kg and turns both switches off. The ledger lives in memory. Restarting the server also starts again from 500 kg.

4. The one job on the desk

There is one sample job. Every screen, every test, and every later file uses these values. Pieces and kilograms are never added together.

Name	Value	Meaning
Company	Motaxle Demo Company	Shown on the Work order screen. Not created by this program.
Work order	WO-2401	The only order the connector will post against.
Finished item	CR-100	Cylinder rod. 100 pieces. Unit is Nos.
Raw item	ST-20	Steel bar. 1.20 kg each, so the issue is 120 kg. Unit is Kg.
Heat	H-441	Batch on the steel. Opening stock is 500 kg in Raw Material Store.
Issue	J-100	120 kg from Raw Material Store to Forging WIP. Raw becomes 380. WIP becomes 120.
Return	R-1	4 kg back against J-100. Raw becomes 384. WIP becomes 116. A second R-1 does not make 8 kg.
Operations	10 and 20	Cutting and Forging. A confirmation stores the numbers and does not receive stock.
Operation 30	Heat treatment	Vendor work. The connector ignores it. It is not an issue message.
Results	96, 2, 1, 1	96 good, 2 rejected, 1 rework, 1 scrap. That is 100 pieces.
Good receipts	C-1 and C-2	40 pieces batch B-1, then 56 pieces batch B-2. Both go to Final QC Hold.
Finished goods	0	Stays 0. Quality releases the hold inside ERPNext, not by a Jinchen message.

Name	Value	Meaning
CAPA	0	The connector must not open a CAPA row.

Exact text matters. ST20 is not ST-20. H441 is not H-441. CR100 is not CR-100. A near miss is refused and stock does not move.

5. Warehouses the stand-in knows

Warehouse	What moves there in this POC
Raw Material Store	Opening 500 kg of ST-20, heat H-441. Issue leaves. Return comes back.
Forging WIP	Receives the 120 kg issue. The 4 kg return leaves it.
Final QC Hold	Good pieces C-1 and C-2. Inspection is required, so a normal receipt lands here.
Finished Goods Store	Stays 0. Asking a receipt to go here returns E-QC and posts nothing.
Rejection Store	2 pieces, reason forge-crack.
Rework Store	1 piece, reason rework-bend.
Scrap Yard	1 piece, reason scale. Scrap is a disposition, not a return of steel.

These names are not used for floor messages in this POC: Job Work Out, Gate Pass Out, Dispatch Bay, Tool Room. The other ERPNext proof uses WIP - Main and item HC-100. Leave that stock alone. Do not rename ST-20 to match it.

6. How one message is decided

Connector.process in connector/poc.py does the work. The desk does not keep a second copy of the rules.

- Test mode refuses to start when a live password is set. The sentence is: Test mode refuses to start when a live password is set. demo.py proves that on startup, then continues with an empty password.
- A message needs id and work_order. If either is missing the code is E-QTY and nothing is posted.
- Unknown kind is E-ITEM.
- For issue, return, disposition, and receipt, the connector looks up the saved stock entry by message id plus work order before it submits a new one. If that entry exists, the answer is E-DUP, status Posted, and the original document name. No second stock entry is created.
- The lookup uses the stock entry, not the log. The Issue J-100 button passes drop_log. The stock entry STE-0001 is saved, the log row is skipped, and a second J-100 still returns E-DUP with raw still 380.
- A confirmation is not a stock entry, so it is not part of that repeat check. Confirming cutting does not receive CR-100.
- Hold, return, scrap, and receipt do not share one another's keys. R-1 does not block C-1.
- Business Rejected does not retry. This POC does not build E-AUTH or E-TIMEOUT. It proves the live-password refusal instead.

7. The five message kinds already coded

Issue

Item must be ST-20, unit Kg, heat H-441, from Raw Material Store to Forging WIP, quantity above zero and not above the free kilograms of that heat. A good J-100 submits one stock entry, takes 120 kg from raw, and adds 120 kg to WIP.

Return

as_scrap is E-REASON before the issued quantity is checked. Scrap is not a return. The return must name an issue that was actually posted, on the same work order. Item and heat must be ST-20 and H-441. Quantity must be above zero and not above the kilograms still out. R-1 moves 4 kg from Forging WIP back to Raw Material Store.

Confirm

Operation 30 returns Posted and the message says it is vendor work and is ignored. It is not stored as a floor confirmation. Operation 10 and 20 store completed, rejected, rework, and scrap. Finished goods stays 0. Any other operation is E-OP. A negative number is E-QTY. Completed above 100 is E-QTY.

Disposition

The reason must match the decision. forge-crack goes with rejection, rework-bend with rework, scale with scrap. Anything else is E-REASON. Quantity must be above zero. The piece count of CR-100 moves into Rejection Store, Rework Store, or Scrap Yard. CAPA stays 0. A disposition does not change the steel kilograms.

Receipt

Item must be CR-100, quantity above zero, batch not blank. If the message asks for Finished Goods Store, the code is E-QC and nothing is posted. Otherwise, because inspection is required, the pieces go to Final QC Hold. C-1 stores batch B-1. C-2 stores batch B-2. Sending C-1 again returns the first document and hold stays 96, not 136.

8. Codes the stand-in already returns

Code	When this POC returns it	Stock
E-DUP	Same id and work order already has a stock entry.	Unchanged. Same document name.
E-ITEM	Unknown kind, item is not ST-20 on an issue, receipt item is not CR-100, or the return has no posted issue.	Unchanged.
E-UOM	Issue unit is not Kg.	Unchanged.
E-WH	Issue is not Raw Material Store to Forging WIP.	Unchanged.
E-BATCH	Heat is not H-441, return is not that heat, or a receipt batch is blank.	Unchanged.
E-STOCK	Issue quantity is above the free kilograms of H-441.	Unchanged.
E-QTY	Missing id or work order, quantity not above zero, return above what is still out, negative confirm, or completed above 100.	Unchanged.
E-OP	Confirm operation is not 10 or 20. Operation 30 is ignored, not E-OP.	Unchanged.
E-REASON	Return flagged as scrap, or disposition reason does not match the decision.	Unchanged.
E-QC	Receipt asks for Finished Goods Store.	Hold and finished stay as they were.

The wider plan also names E-BOM, E-CLOSED, E-SERIAL, E-AMEND, E-AUTH, and E-TIMEOUT. Those are not built in this stand-in. Do not add them in order to make the current demo pass.

9. What the tiles mean after a full walk

After	Raw kg	WIP kg	Hold	Finished	Rej	Rework	Scrap	Entries
Open	500	0	0	0	0	0	0	0
J-100, log dropped	380	120	0	0	0	0	0	1
J-100 again	380	120	0	0	0	0	0	1
R-1	384	116	0	0	0	0	0	2
R-1 again	384	116	0	0	0	0	0	2
Confirm 10 and 20	384	116	0	0	0	0	0	2
Reject, rework, scrap	384	116	0	0	2	1	1	5
C-1 and C-2	384	116	96	0	2	1	1	7
C-1 again and E-QC	384	116	96	0	2	1	1	7

CAPA is 0 on every row. Pieces accounted at the end are 96 + 2 + 1 + 1 = 100. Stock documents are named STE-0001 upward. The first issue is STE-0001 even when the log write is dropped.

10. Heat lookup

Trace asks for a rod batch, not for a typed heat. After C-1, lookup of B-1 returns found, work order WO-2401, heat H-441, and the stock document of that receipt. That link exists only if an issue of heat H-441 was posted on the same work order. Lookup of H441 returns missing. The desk does not invent a heat and it does not treat a missing hyphen as the same batch. After C-2, B-2 is found the same way.

11. The website, screen by screen

server.py serves one page. The left side is the navigation. The top strip is seven tiles: raw kg, WIP kg, hold, finished, rejection, rework, scrap. A highlighted button is the suggested next step. Suggestion follows posted documents and stored confirmations, not a click that was only queued.

Screen	What is already on it
Today	WO-2401, the route, issued and returned kilograms, pieces accounted out of 100, finished goods, CAPA, document count, and the piece bar. Switch A, Switch B, and Release WO-2401 sit here.
Work order	Company, CR-100, 100 pieces, BOM 1.20 kg, heat H-441, issued, returned, WIP, and the three operations.

Screen	What is already on it
Material	Issue J-100, send it again, wrong item, wrong heat, 999 kg, unit Nos, issue from finished goods, return as scrap, return R-1, send R-1 again. Plus the form to type your own message.
Operations	Confirm cutting, confirm forging, confirm heat treatment, confirm operation 40, confirm 150 pieces. Finished goods does not change here.
Quality	Reject 2, rework 1, scrap 1, receive B-1, receive B-2, send C-1 again, put pieces in finished goods, receive with a blank batch.
Trace	Look up a batch. Buttons for B-1, B-2, and H441.
Message log	Every attempt, with a filter, and the stock documents underneath.

Send your own message is on Material, Operations, and Quality. The work order is fixed as WO-2401. You type the id and the fields for that kind. The same Connector.process handles it. A repeated J-100 typed by hand is still E-DUP.

12. Switches and release, already on Today

- Switch A, when on, accepts issue, return, disposition, and receipt as Queued. It writes the attempt and does not call process. Raw stays 500 if you queue the issue before anything else. Turning Switch A off does not post that queue.
- A confirmation is not held by Switch A. The stock kinds are.
- Switch B, when on, blocks Release WO-2401. The log says it was not released. No stock entry is created either way.
- Switch B off: Release records that WO-2401 was released on this desk and that Jinchen was not called. It does not create a stock entry and it does not receive finished goods.
- Reset asks for confirmation, then returns to 500 kg, switches off, not released, empty log.

13. What demo.py prints

Run it once before changing code. The last line says it did not call Jinchen and it did not call a live ERPNext site. The walk is:

- Startup with a live password is refused.
- Opening line: 500 kg is already in ERPNext. Jinchen did not receive it.
- J-100 posts, then the log write fails. Document STE-0001. Raw 380, WIP 120, one entry.
- The same J-100 again is E-DUP and the same document. Raw stays 380.
- ST20 is E-ITEM. H441 is E-BATCH. Raw stays 380.
- R-1 posts. The same R-1 again does not make 8 kg. Raw 384, WIP 116, two entries.
- Cutting confirmed. Finished goods stays 0.
- Rejection 2, rework 1, scrap 1. CAPA 0.
- C-1 and C-2, then C-1 again is E-DUP. Asking finished goods for the rejected pieces is E-QC. Hold 96, finished 0, seven entries.
- Lookup B-1 is found, heat H-441, work order WO-2401. Lookup H441 is missing.

14. The twelve tests already in the folder

Test	What a pass means
tc02 live password	Test mode with a password raises ConfigError. An empty password still constructs the connector.
tc03 issue once	One issue. Raw 380. WIP 120.
tc04 repeat after log loss	drop_log leaves the log empty, the entry saved, and the second call E-DUP on the same document.
tc05 near miss	ST20 is E-ITEM. H441 is E-BATCH. Raw stays 500. No entry.
tc06 return once	R-1 twice leaves raw 384 and WIP 116.
tc07 confirm	Cutting posts no CR-100. Finished and hold stay 0.
tc08 disposition	2, 1, and 1. Finished 0. CAPA 0.
tc09 receipts	C-1, C-2, and C-1 again. Hold 96. Finished 0.
tc10 E-QC	A receipt aimed at finished goods does not land there.
tc11 lookup	B-1 finds H-441 and WO-2401. H441 is missing.
scrap return	as_scrap is E-REASON. Raw stays 380 after the issue.
operation 30	The message says ignored. Finished stays 0.

A pass prints Ran 12 tests, OK. Do not edit a test to change 380, 384, 96, or 0. If a tile on the desk disagrees with this table, stop and copy the row. Do not post a correcting message.

15. What this POC is deliberately not

- Not a Jinchen screen and not a Jinchen HTTP call.
- Not the Motaxle ERPNext app, and not a new DocType.
- Not purchase, incoming inspection, RGP or NRGP, tool life, packing, dispatch, invoice, GST, calibration, maintenance, opening accounts, or CAPA.
- Not a second stock system. When the test site is ready, ERPNext remains the place that posts stock. This ledger is only the local proof.
- M6 confirmation must not receive finished goods. M7 receipt is a different message, and in this stand-in that receipt stops in Final QC Hold.

16. How to use this when you start the tasks

The pages after this guide are the remaining work. Section numbers there are unchanged. Both of you walk the desk in section 7 before new code, and the tiles you write down must match section 9 of this guide. Ashish adds files and an Inbox screen that call the process function already in connector/poc.py. He does not recheck ST-20 inside the reader. Ananya adds the checks and the proof script around the same function. Part B, the ERPNext site, stays closed until the lead sends the site URL, the company, the stock entry type name, and a user that cannot submit a sales invoice.

All remaining POC work, split in half

The lead pushes motaxle-jinchen-connector and sends one clone address. This file is everything still left for the POC demo. Part A is local and starts now. Part B is the same story on the ERPNext test site, and it stays closed until the lead writes that the site is ready. Ashish and Ananya each have the same number of tasks in Part A, and the same number in Part B.

Both read sections 1 to 6. Both do section 7 before any new code. Ashish does section 8 (A1 to A9). Ananya does section 9 (N1 to N9). Section 10 is the joint sitting. Section 11 is Part B. Section 12 is the note. Section 13 is the stop line.

1. Already in the repository. Do not rebuild it

Already done	Where	Leave it
Ledger, repeat rule, failed log write on the J-100 button	connector/poc.py and the Material screen	Do not write a second ledger.
Printed story and 12 tests	demo.py, tests/test_poc.py	Do not change 380, 384, 96, or 0 to force a pass.
Desk screens and Send your own message	server.py	Do not remove Today, Work order, Material, Operations, Quality, Trace, or Message log.
Switch A, Switch B, Release	server.py	A file must use this switch. Do not add a second switch.

Job values, used everywhere: Motaxle Demo Company, WO-2401, CR-100, 100 pieces, ST-20 at 1.20 kg, heat H-441, opening 500 kg in Raw Material Store. Issue J-100 is 120 kg. Return R-1 is 4 kg. Results are 96 good, 2 rejected, 1 rework, 1 scrap. Receipts C-1 = 40 batch B-1 and C-2 = 56 batch B-2, both to Final QC Hold. Operations 10 Cutting and 20 Forging. Operation 30 Heat treatment is vendor work and is ignored. Finished goods stays 0. CAPA stays 0. Pieces and kilograms are never added. ST20 is not ST-20. H441 is not H-441.

2. Everything still open

Gap	Part	Who
Walk the desk that already runs, and record the tiles	A, before new code	Both, section 7
Sample JSON files, inbox reader, Inbox screen	A	Ashish A2 to A4
Switch A on a file: 500, then 500, then 380	A	Ashish A5
Test mode refuses a live password from a config file	A	Ashish A7
First J-100 file still posts when the log write is dropped	A	Ashish A8
README for run, test, inbox, and switch	A	Ashish A9
Refusal tests that do not exist yet, including a repeated rejection	A	Ananya N2 and N7
Bad example files on disk	A	Ananya N3
prove.py prints TC-02 to TC-11	A	Ananya N4
Switch B, batch B-2, and 100 pieces accounted	A	Ananya N5
Spoken demo script and the local screenshot set	A	Ananya N8 and N9
Same numbers as real bins on the test site	B, gated	Ashish B1 to B4, Ananya B1 to B4

3. Out of this POC

- No Jinchen web address, token, or path from another factory. The API pack is not signed.
- No CAPA, invoice, purchase, dispatch, packing, tool life, or feasibility study.
- No second repository, no bench new-app, no copy of this folder into motaxle-bench/apps, no new DocType.
- No new Python package. Python 3.9 or newer and the standard library only.
- No HC-100, and no rename of ST-20. Do not post this job into a live company.
- Part B does not start on a guess. If the site URL or the stock entry type name is missing, stop that part and leave the gap in the note.

4. One repository

```
git clone THE_ADDRESS_THE_LEAD_SENDS
cd motaxle-jinchen-connector
python3 -m unittest tests/test_poc.py
git checkout -b ashish-inbox
git checkout -b ananya-checks
```

Ashish uses only ashish-inbox. Ananya uses only ananya-checks. Do not commit on the default branch. Do not force-push. Push the branch and tell the lead. The lead merges. A commit message says why. Do not commit a password, a .env file, or a live site URL. inbox/in stays empty in git. inbox/results from a local run are not committed.

5. Ledger you must hit

After	Raw	WIP	Hold	Finished	Rej.	Rework	Scrap	CAPA	Entries
Opening	500	0	0	0	0	0	0	0	0
J-100, log write dropped	380	120	0	0	0	0	0	0	1
J-100 again	380	120	0	0	0	0	0	0	1
R-1	384	116	0	0	0	0	0	0	2
R-1 again	384	116	0	0	0	0	0	0	2
Confirm 10 and 20	384	116	0	0	0	0	0	0	2
Disposition	384	116	0	0	2	1	1	0	5
Same rejection id again	384	116	0	0	2	1	1	0	5
C-1 and C-2	384	116	96	0	2	1	1	0	7
C-1 again and E-QC	384	116	96	0	2	1	1	0	7

Repeat key is the message id plus the work order, searched on the saved stock entry. A missing log row does not allow a second post. Hold, return, scrap, and receipt do not share one another's keys.

6. Files and fields

Path	Who	Rule
inbox/samples/	Ashish	Good files below. Reading the samples folder itself does not post.
inbox/in/	Ashish	Empty in git. Filled only by Copy samples or a test.
inbox/done/ and inbox/results/	Ashish	Moved and written only after process returns. Result JSON has status, code, document, message.
connector/inbox.py	Ashish	Sorts by filename, calls the existing process. No item or heat check of its own.
config.json	Ashish	mode is test. live_password is an empty string in git.
README.md	Ashish, prove lines from Ananya	How to clone, test, prove, start the desk, and read the inbox.
inbox/examples/	Ananya	Bad files. The happy-path button must not copy this folder.
prove.py	Ananya	Prints TC-02 to TC-11. Does not open a port and does not read inbox/in.
demo/SCRIPT.txt and demo/shots/	Ananya	The words to say, and the local screenshots. No password in a shot.

Good files in inbox/samples. Quantity values are numbers. 08 and 09 do not set to_warehouse.

File	Fields
01-issue-J-100-log-fails.json	kind issue, id J-100, work_order WO-2401, item ST-20, qty 120, uom Kg, batch H-441, from_warehouse Raw Material Store, to_warehouse Forging WIP. The name log-fails tells the reader to drop the log write.
02-return-R-1.json	kind return, id R-1, work_order WO-2401, against J-100, item ST-20, qty 4, uom Kg, batch H-441
03-confirm-op10.json	kind confirm, id CF-10, work_order WO-2401, operation 10, completed 100, rejected 0, rework 0, scrap 0
04-confirm-op20.json	kind confirm, id CF-20, work_order WO-2401, operation 20, completed 100, rejected 0, rework 0, scrap 0
05-reject-2.json	kind disposition, id D-REJ, work_order WO-2401, item CR-100, qty 2, decision rejection, reason forge-crack
06-rework-1.json	kind disposition, id D-RW, work_order WO-2401, item CR-100, qty 1, decision rework, reason rework-bend
07-scrap-1.json	kind disposition, id D-SC, work_order WO-2401, item CR-100, qty 1, decision scrap, reason scale
08-receipt-C-1.json	kind receipt, id C-1, work_order WO-2401, item CR-100, qty 40, uom Nos, batch B-1
09-receipt-C-2.json	kind receipt, id C-2, work_order WO-2401, item CR-100, qty 56, uom Nos, batch B-2

Bad files in inbox/examples. Ananya writes them. Each one leaves stock where it was.

File	Wrong field	Code
ex-st20.json	item ST20 on a 1 kg issue of heat H-441	E-ITEM
ex-h441.json	batch H441, item ST-20, qty 1	E-BATCH
ex-nos.json	uom Nos	E-UOM
ex-warehouse.json	from_warehouse Finished Goods Store	E-WH
ex-999.json	qty 999 of ST-20 heat H-441	E-STOCK
ex-finished.json	receipt C-BAD, CR-100, qty 2, batch B-X, to_warehouse Finished Goods Store	E-QC

File	Wrong field	Code
ex-nobatch.json	receipt C-NONE, CR-100, qty 1, batch empty	E-BATCH
ex-scrap-return.json	return R-SCRAP against J-100, qty 1, as_scrap true. Needs J-100 already posted for the stock figure, but the code is E-REASON even before that.	E-REASON

7. Both of you, before new code

P1 Walk the desk that already runs

Do this

- From the cloned folder run python3 -m unittest tests/test_poc.py and python3 demo.py.
- Run python3 server.py and open http://127.0.0.1:8765. The port is required.
- Reset job. Send Issue J-100. Send it again. Send Return R-1 and send it again. Confirm cutting and forging. Reject 2, rework 1, scrap 1. Receive B-1 and B-2. Send C-1 again. Send the finished-goods receipt. Look up B-1 and H441.
- Each person writes the tiles after J-100, after the repeat, after R-1, after cutting, and after the receipts. Then Reset job so the next person starts from 500.

Expected

Tests say Ran 12 tests, OK. demo.py says it did not call Jinchen. Tiles match section 5: raw 380 then 384, hold 96, finished 0, entries 7, CAPA 0. B-1 finds WO-2401 and H-441. H441 is missing.

If it fails

A wrong tile stops you. Copy the row. Do not edit the test and do not post a correcting message. A blank page means the address is missing :8765.

Finished when

Both notes contain those tiles, and the desk is reset to 500 kg.

8. Ashish, branch ashish-inbox

Nine tasks. Do not edit prove.py. Do not recheck ST-20 or H-441 inside the reader.

A1 Baseline and branch

Do this

- P1 is done. Create ashish-inbox from that clean point.
- Do not commit on the default branch.

Expected

git status shows ashish-inbox. The 12 tests still pass.

If it fails

If P1 was red, do not branch. Send the failing test name.

Finished when

The branch exists and no new file is committed yet.

A2 Nine sample files

Do this

- Create inbox/samples, inbox/in, inbox/done, and inbox/results. Put a file named KEEP in each empty folder so git keeps it.
- Write the nine JSON files from section 6 into inbox/samples only. Two-space indent. No extra fields.
- Do not copy them into inbox/in in the commit.

Expected

01-issue-J-100-log-fails.json has ST-20, H-441, and qty 120 as a number. 08 and 09 have no to_warehouse. inbox/in has no message file.

If it fails

If a field name is unclear, copy it from demo.py. Do not rename id.

Finished when

The nine samples are on the branch.

A3 Reader

Do this

- Add connector/inbox.py. List inbox/in by filename.
- read_next(connector) parses the first file and calls connector.process. If the file name contains log-fails, pass drop_log=True. Write inbox/results/.json with status, code, document, and message. Move the source to inbox/done.
- If the file is not a JSON object, do not call process. Write code E-ITEM and move it to inbox/done. Stock stays put.
- read_all(connector, switch_a) calls read_next until inbox/in is empty. When switch_a is true, do not call process and do not move the file. Write status Queued. Turning the flag to false does not call read_all by itself.

Expected

A test uses a temporary folder, copies the issue sample, and gets raw 380 and WIP 120 even though the log write was dropped. A second file with id J-100 returns E-DUP and raw stays 380. The sample in inbox/samples is still there.

If it fails

A second stock calculation inside inbox.py is a stop. Delete it. If the second read creates another issue document, the repeat rule is broken.

Finished when

The reader tests pass and tests/test_poc.py still says OK.

A4 Inbox screen

Do this

- Add a desk screen named Inbox. Keep every existing screen.
- It lists file names in inbox/in.
- Copy samples copies inbox/samples into inbox/in only when inbox/in has no json files. Otherwise it says the inbox is not empty and copies nothing.
- Read inbox calls read_all with the current Switch A flag, through the same Desk path the buttons use, so the switch cannot be skipped.
- Stay on 127.0.0.1 port 8765.

Expected

Reset, copy samples, read inbox. Tiles match the last row of section 5. inbox/in is empty. inbox/done has nine files. Trace B-1 finds H-441 and WO-2401. The first issue did not need a log row to block the repeat.

If it fails

Wrong order means the file names, not the ledger. Hold 136 means C-1 was read twice. Stop and name the duplicate.

Finished when

A clean reset can be demoed from files alone, and the Material J-100 button still works afterwards.

A5 Switch A on a file

Do this

- Reset the ledger. Empty inbox/in and inbox/done for this test.
- Switch A on. Copy only the issue sample into inbox/in. Read inbox.
- Switch A off. Do not read yet.
- Read inbox once.

Expected

On: file stays in inbox/in, status Queued, raw 500. Off, before the next read: raw still 500 and the file still waiting. After the read: raw 380, WIP 120, file in inbox/done.

If it fails

If the off click itself moves raw, remove that flush.

Finished when

A test asserts raw 500, then 500, then 380.

A6 Old desk still posts

Do this

- Run python3 -m unittest discover tests and python3 demo.py.
- Reset, then use the Material Issue J-100 button once.

Expected

Discover and the 12 tests are OK. demo.py numbers are unchanged. The button reaches raw 380 and one document.

If it fails

Do not delete a screen to make a test pass.

Finished when

You have not pushed yet. A7 to A9 are still yours.

A7 Test configuration

Do this

- Add config.json with mode set to test and live_password set to an empty string.
- server.py, prove.py once it exists, and read_all refuse to start when mode is test and live_password is not empty. They also refuse when mode is not test or live.
- The refusal is the same sentence the code already uses: test mode refuses to start when a live password is set.
- A test points at a temporary config. Do not commit a non-empty password. Do not print the password in a result file.

Expected

Empty password: the desk starts. A temporary config with any non-empty live password: the process stops and posts nothing. git shows live_password as an empty string.

If it fails

If the server starts with a filled password, stop. Remove the path that ignores config.json.

Finished when

The committed config has an empty live password, and a test proves the refusal.

A8 Dropped log write from a file

Do this

- The sample name contains log-fails. read_next passes drop_log=True only for that file.
- After that read, one stock entry exists for J-100 and the connector log has no row for that id.
- Copy a second file with the same id and work order, whose name does not contain log-fails. Read it.

Expected

First read: raw 380, WIP 120, one entry, no log row for J-100. Second read: E-DUP, same document name, raw 380, still one issue entry.

If it fails

If the first read does not create the entry, drop_log was applied to the stock save. It applies only to the log row. If the second read creates STE-0002, stop.

Finished when

A test covers both reads. The other eight samples do not use drop_log.

A9 README and push

Do this

- README.md tells a new person: clone, python3 -m unittest discover tests, python3 demo.py, python3 prove.py, python3 server.py, then http://127.0.0.1:8765.
- It says Copy samples then Read inbox, and it says Switch A does not flush when turned off.
- It says the JSON fields are the POC contract, not a signed Jinchen API.
- Leave a short heading Prove for Ananya's sentence. Do not delete it.
- Commit your files only. Push ashish-inbox. Tell the lead the branch name.

Expected

A person who has the clone address can follow the README without asking you. The branch is on the shared repository. The default branch is untouched by you.

If it fails

Push rejected: send the error text. Do not force-push.

Finished when

The lead can open ashish-inbox, and A3, A5, A7, and A8 tests pass there.

9. Ananya, branch ananya-checks

Nine tasks. You do not need the inbox reader for N2 to N8. Do not rewrite connector/poc.py to make a test pass. Do not drop bad files into inbox/in.

N1 Baseline and branch

Do this

- P1 is done on the same clone address. Do not create a repository.
- Create ananya-checks.

Expected

The 12 tests pass on ananya-checks.

If it fails

A red baseline stops N2. Send the test name.

Finished when

The branch exists and you have not edited poc.py.

N2 Refusal tests that are missing

Do this

- Add tests/test_checks.py. One fresh Connector per test.
- Cover E-UOM, E-WH, E-STOCK with qty 999, E-OP with operation 40, E-QTY with completed 150, E-BATCH with an empty receipt batch, and E-REASON with decision rejection and reason scale.
- Also cover as_scrap on a return: E-REASON.
- Raw stays 500 when you have not issued. Finished stays 0. capa stays 0.

Expected

python3 -m unittest tests/test_checks.py is OK, and the original 12 tests are still OK.

If it fails

A moved kilogram with an empty code is a product gap. Do not edit the expected code to match it. Send the message and the code.

Finished when

test_checks.py is on the branch.

N3 Bad files on disk

Do this

- Create inbox/examples and the eight files in section 6.
- A test loads ex-st20.json and ex-finished.json from disk and passes the object to Connector.process.
- No button and no Copy samples path may read inbox/examples.

Expected

ex-st20.json returns E-ITEM and raw stays 500. ex-finished.json returns E-QC and finished stays 0.

If it fails

A parse error is a comma. Do not change ST20 to ST-20.

Finished when

The eight files and the two disk tests are committed.

N4 prove.py

Do this

- Add prove.py at the repository root. It uses Connector and require_test_mode. It does not bind a port and does not read inbox/in.
- TC-02: require_test_mode('test', 'live-secret') raises, and Connector() with an empty live password starts.
- Then the section 5 story. The first J-100 uses drop_log=True. Then the same J-100 again.
- Include R-1 twice, confirm 10, confirm 20, the three dispositions, the same D-REJ again, C-1, C-2, C-1 again, and a receipt to Finished Goods Store.
- Print TC-02 through TC-11, each ending PASS or FAIL, then ALL PASS or NOT PASS. Also print B-2 found with heat H-441.
- Under the Prove heading in README.md, add one sentence: python3 prove.py prints ALL PASS.

Expected

python3 prove.py prints ALL PASS. Hold on the receipt case is 96. The repeated rejection leaves rejection at 2, not 4. Entries at the end are 7. CAPA is 0. Finished is 0.

If it fails

PASS without a number check is not a proof. A FAIL must not be swallowed into ALL PASS.

Finished when

Someone who has not read the code can run one command and see the eleven lines.

N5 Switch B, B-2, and the 100 pieces

Do this

- Add tests/test_desk.py. Import Desk from server. Do not call main.
- Switch B on, then release. released is false. The message says it was not released. Entries stay 0.
- Switch B off, then release. released is true. The message says Jinchen was not called. Entries stay 0.
- After J-100, C-1, and C-2, lookup B-2 is found with heat H-441 and work order WO-2401. H441 is missing.
- hold + rejection + rework + scrap is 100 and finished is 0.

Expected

python3 -m unittest tests/test_desk.py is OK. Release creates no stock entry.

If it fails

If import starts port 8765, main was called. Stop the old server with Ctrl+C. Do not change the port.

Finished when

The desk tests pass without a browser.

N6 The remaining refusals

Do this

- Extend tests/test_checks.py with one test each.
- Missing id or missing work order: E-QTY, no entry.
- Unknown kind: E-ITEM, no entry.
- Return R-X against J-100 before any issue exists: E-ITEM, raw 500.
- After one J-100, a return of 999 kg: E-QTY, raw stays 380.
- After one J-100, a return with batch H441: E-BATCH, raw stays 380.
- Confirm with completed -1: E-QTY, finished 0.
- Disposition qty 0: E-QTY, capa 0.
- Receipt item ST-20: E-ITEM, finished 0.
- Receipt qty 0: E-QTY, hold 0.
- D-REJ posted once, then the same id again: E-DUP, rejection stays 2.

Expected

Every row above is a passing test. Stock matches the stock column. capa stays 0.

If it fails

Do not delete a rule in poc.py because a test is awkward. Send the lead the row that disagrees.

Finished when

test_checks.py covers this table and N2, and discover is OK on your branch before N8.

N7 Demo script

Do this

- Add demo/SCRIPT.txt. It is the words for an 8 minute walk. Use these lines, in this order.
- 0:00 This program is not Jinchen and it is not the Motaxle app. The file is a stand-in because the API pack is not signed.

- 0:40 Five hundred kilograms opened in ERPNext. Jinchen did not receive them.
- 1:20 One issue. Raw store 380. WIP 120. One stock entry, heat H-441.
- 2:20 Sent twice. The raw store is still 380. That is the proof.
- 3:10 Four kilograms came back. Raw 384. WIP 116. The same return does not make 8 kilograms.
- 4:00 Cutting is confirmed. Finished goods is still zero. Confirmation is not a receipt.
- 4:50 Two rejected, one rework, one scrap. Finished goods is still zero. No CAPA was opened.
- 5:50 Forty plus fifty-six is ninety-six. They sit in Final QC Hold. Finished goods is still zero.
- 6:40 The hold is still 96, not 136. Rejected pieces cannot enter finished goods.
- 7:20 Batch B-1 reads back to H-441 and WO-2401. H441 does not match.
- 8:00 The ERPNext POC proved the business flow. This one proved the connector will not double the metal. The real Jinchen call waits for the API document.
- Under each line, name the screen: Today, Material, Operations, Quality, or Trace.

Expected

demo/SCRIPT.txt has all eleven lines and a screen name on each. It does not tell anyone to call Jinchen.

If it fails

Do not shorten 380 to about 400. The figures are the proof.

Finished when

The file is on ananya-checks.

N8 Local screenshots

Do this

- Create demo/shots. Reset the desk and capture only these. No password, no browser profile bar if it shows an account.
- Opening tiles at raw 500.
- After J-100: raw 380, WIP 120, and the line that the log write failed.
- The repeat: E-DUP and the same document name, raw still 380.
- Finished goods 0 after cutting.
- Hold 96, rejection 2, rework 1, scrap 1, CAPA 0.
- Trace for B-1 and for H441.
- Switch A on, raw still 500, status Queued. If Ashish's Inbox screen is not merged yet, shoot the Material button with Switch A on. Say which one you shot.
- Terminal line from prove.py or from a one-off call that test mode refuses a live password. Then remove any temp password before you commit.

Expected

Eight pictures, file names that match the list. Finished goods reads 0 in the cutting shot. H441 reads missing.

If it fails

If a shot shows a password, delete the file. Do not commit it.

Finished when

demo/shots is on the branch and contains no secret.

N9 Push

Do this

- Run python3 -m unittest discover tests, python3 demo.py, and python3 prove.py.
- Commit tests, inbox/examples, prove.py, demo/SCRIPT.txt, demo/shots, and the Prove sentence in README.md.
- Do not commit inbox/in messages or inbox/results. Push ananya-checks and send the branch name.

Expected

discover is OK. prove.py prints ALL PASS. The branch is on the shared repository. The default branch is still the lead's.

If it fails

If discover runs a test from ashish-inbox, you are on the wrong branch. Do not delete his files.

Finished when

The lead can see ananya-checks.

10. Together

T1 Both halves on one desk

Do this

- Wait until the lead has merged both branches, or use the merge the lead names. Do not rewrite the other commit.
- Run discover, demo.py, and prove.py.

- Reset. Copy samples. Read inbox. Compare with the last rows of section 5.
- Reset. Switch A on. Copy only the issue sample. Read inbox. Raw 500. Switch A off. Raw still 500. Read inbox. Raw 380.
- Reset. Copy ex-st20.json into inbox/in by hand. Read inbox. E-ITEM. Raw 500.
- After a full sample read, Trace B-1 and H441.
- Read demo/SCRIPT.txt aloud once against that desk. Stop at 8 minutes. Do not skip the 380 line.

Expected

ALL PASS. Nine-file read matches section 5. Switch A is 500, 500, 380. ex-st20 does not move stock. B-1 finds H-441. H441 is missing. The Material J-100 button still works after a reset.

If it fails

A wrong tile ends the sitting. Write the row. Do not type a fixing stock entry. Do not force-push.

Finished when

The section 12 note is filled from this sitting.

11. Part B. Do not start until the lead says the test site is ready

Part A is the demo you can do now. Part B is the same job on the ERPNext test site. Open it only when the lead sends all four of these: the site URL, the company on that site, the stock entry type name for an issue, and a user that cannot submit a sales invoice. If any one is missing, write the gap and stop. Do not invent the type name. Do not install ERPNext. Do not type a stock entry by hand to hide the gap. Do not point the connector at a live company.

Ashish, Part B

B1 Open only the site the lead named

Do this

- Use the URL, the company, and the user from the lead's message.
- If login fails, stop. Do not try the Administrator password from a guide, and do not send a password back.

Expected

You are on Motaxle Demo Company, or the company name the lead wrote if it differs. You can say which name you saw.

If it fails

Wrong company: stop. Do not create a second company.

Finished when

The note says the company name and that you did not create one.

B2 Warehouses and opening metal

Do this

- Check these seven names: Raw Material Store, Forging WIP, Final QC Hold, Finished Goods Store, Rejection Store, Rework Store, Scrap Yard.
- If a name is missing, stop. Do not create a near name and do not rename Work In Progress - Main.
- ST-20 must be batch tracked. If it already exists, reuse it. Do not create a second ST-20.
- Receive 500 kg of batch H-441 into Raw Material Store with ERPNext's own material receipt. The connector is not used for this receipt.

Expected

The bin shows 500 kg of H-441. Finished goods for CR-100 is 0. The connector log has no row for this receipt.

If it fails

If the receipt screen will not take a batch, stop and say so. Do not turn batch tracking off to force the save.

Finished when

The note has the bin figure 500 and the list of any missing warehouse.

B3 BOM and released work order

Do this

- Submit a BOM for CR-100: ST-20, 1.20 kg each. Operations 10 Cutting and 20 Forging. Operation 30 Heat treatment marked as vendor work and not a step the connector will confirm.
- Release WO-2401 for 100 Nos of CR-100 against that BOM. If WO-2401 already exists, reuse it. Do not create WO-2401-2.

Expected

The work order is released. Operation 30 is visible as vendor work. No issue has been posted yet. Raw is still 500.

If it fails

If the BOM will not submit, stop. Do not delete a BOM that belongs to HC-100.

Finished when

The note says WO-2401 is released and raw is still 500.

B4 One issue, only if the type name was given

Do this

- Read the stock entry type name the lead sent. If the message does not contain it, stop this task.
- Post J-100 once, 120 kg, ST-20, batch H-441, from Raw Material Store to Forging WIP, against WO-2401, using that type.
- Post the same J-100 again through the connector path you already trust. Do not click a second manual entry to see what happens.

Expected

One stock entry. Raw 380. WIP 120. The second attempt does not create a second entry. The document name on the repeat is the first document.

If it fails

If you do not know the API body, stop. Do not guess fields. Write API body not provided. The local prove.py result still stands.

Finished when

The note says either the 380 figure and the document name, or the exact missing item that stopped you.

Ananya, Part B

B1 Finished item and a zero finished bin

Do this

- On the same site and company, CR-100 is in Nos. Reuse it if it exists.
- Before anyone receipts the rod, the finished-goods bin for CR-100 is 0.
- Do not create a second CR-100. Do not create a stock entry type.

Expected

You can point at CR-100 and at a finished bin of 0. You did not add a warehouse.

If it fails

If CR-100 is missing, stop and tell the lead. Do not invent a code such as CR100.

Finished when

The note says finished goods for CR-100 was 0 before any receipt.

B2 Read the bins. Do not post

Do this

- After Ashish's opening receipt, read the H-441 bin. Write the number.
- After his J-100, if B4 was not stopped, read raw and WIP. Write both.
- If B4 stopped, write the gap he wrote. Do not post the issue yourself.

Expected

The numbers are 500 before the issue, and 380 and 120 after a real issue. They match section 5, or the note says the site step did not run.

If it fails

A hand-typed stock entry to force 380 is a fail. Delete is not your fix. Tell the lead.

Finished when

The note has the bin numbers or the gap sentence, and nothing was posted by you.

B3 Confirmation and hold are still not finished goods

Do this

- If the site has no confirmation yet, do not invent one. Say site confirmation not posted.
- If a confirmation was posted, the finished bin for CR-100 is still 0.
- Good pieces, if any were receipted on the site, are in Final QC Hold. They are not in Finished Goods Store.
- CAPA has no new row from this work.

Expected

Finished stays 0. CAPA has no new row. Hold, if a receipt exists, is the receipt quantity and not double.

If it fails

If finished goods moved because of a confirmation, stop and name the document. Do not reverse it quietly.

Finished when

The note says finished 0 and whether a site receipt exists.

B4 Site screenshots, only of what exists

Do this

- Add shots only for screens you opened: the 500 kg bin, the 380 kg bin if the issue exists, finished goods at 0, and the CAPA list with no new row.
- Name each file with the figure in it, such as bin-500.png. Put them in demo/shots/site.
- Do not photograph a password. If a shot is impossible because the step stopped, do not fake it. List it as not taken and why.

Expected

Every file matches a real screen from this site. The note lists any shot you did not take.

If it fails

A mock image is a fail. Remove it.

Finished when

demo/shots/site is committed only with real screens, or the note explains the gap and the folder is absent.

12. The note

Ashish

- Branch name. Old 12 tests OK. Reader tests OK.
- Raw after a full sample read.
- Switch A file figures: 500, 500, 380.
- Log-fails: one entry, no log row, second file E-DUP, raw still 380.
- config.json live_password is empty, and the refusal test exists.
- Part B: either raw 380 with the document name, or the missing item that stopped B4.

Ananya

- Branch name. discover OK. Last line of prove.py.
- N2 and N6 codes, each with no stock move.
- Repeated rejection stayed at 2.
- B-1 and H441 lines from P1 or T1.
- Part B: finished goods 0, and any shot you did not take.

Joint

- Hold, finished, and stock entry count after the nine files.
- ex-st20 code and raw kilograms.
- That the script was read once and the 380 line was not skipped.

13. Stop line

- Do not add a Jinchen URL.
- Do not flush Switch A by turning it off, and do not let Read inbox run on a timer.
- Do not receive operation 10 into finished goods, and do not open CAPA.
- Do not start Part B without the lead's four items. Do not guess a stock entry type. Do not type a manual entry to hide a gap.
- Do not install ERPNext as part of this file, and do not use a live company.
- This work sits inside the 53 to 75 days. It is not added on top of the ERPNext 120 to 140.

Finished when Part A is merged, prove.py prints ALL PASS, the nine files hit section 5, the script and the local shots exist, and Part B is either matched to those figures or stopped in writing on a named gap. The repository still does not call Jinchen.